Student Hub Platform – Final Project Documentation

Information Technology Management

University of Primorska – FAMNIT

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Team Members

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1. Introduction

The Student Hub Platform project was developed as part of the Information Technology Management course with the goal of creating a centralized academic platform for university students. Throughout the project, the team analyzed student needs, defined system requirements, designed personas, tested hypotheses, prepared implementation plans, and evaluated usability through structured testing.

Modern students often rely on multiple disconnected tools such as Moodle, WhatsApp groups, Google Drive folders, Discord servers, and social media pages to manage academic resources, communication, and university events. This fragmentation creates confusion, inefficiency, and difficulty in organizing study related information.

The Student Hub Platform aims to solve these issues by providing a unified web based system where students can:

- Access study materials
- Track academic progress
- Participate in discussion forums
- Discover and contribute events
- Collaborate with peers
- Manage their academic life in one place

This document summarizes the complete project process, including planning, research, implementation preparation, stakeholder analysis, usability testing, and final conclusions.

2. Project Charter

2.1 Project Code and Name

SHP-2026: Student Hub Platform

2.2 Project Client

Famnit University of Primorska – Student Services Department

2.3 Project Importance

The Student Hub Platform addresses the growing need for centralized access to academic resources and student collaboration.

Students currently experience problems such as:

- Study materials being scattered across group chats and cloud folders
- Difficulty tracking academic progress
- Poor communication between students
- Lack of awareness of university and local events

The project improves the university learning environment by:

- Providing a single access point for study resources
- Encouraging collaboration through forums and file sharing
- Improving transparency of academic progress
- Increasing student engagement in university events

2.4 Project Purpose

The purpose of the project is to develop a web-based platform where students can:

- Access study materials
- Track course and exam progress
- Communicate through forums
- Stay informed about events
- Collaborate with peers

2.5 Main Objectives

The system should:

- Allow student registration and profile personalization
- Support uploading and downloading study materials
- Provide academic progress tracking
- Include discussion forums
- Include an event management system
- Provide administrator moderation tools

3. Project Purpose and Objectives

The main goal of Student Hub Platform is to simplify and improve student academic life through digital centralization.

The project focuses on four key areas:

3.1 Study Material Accessibility

Students often lose time searching for lecture notes and exam materials across multiple communication channels. The platform provides a structured repository organized by:

- Subject
- Academic year
- Faculty
- Study programme

3.2 Academic Progress Tracking

Students need a clearer overview of completed and pending obligations. The platform enables:

- Subject progress tracking
- Exam tracking
- Completion status monitoring
- Progress visualization

3.3 Student Communication

The integrated discussion forum allows students to:

- Ask questions
- Share knowledge
- Help colleagues
- Discuss course-related topics

3.4 Event Awareness

Students often miss important academic or social events. The event tracker enables:

- Viewing events
- Filtering events
- Event reminders
- Student event contributions

4. Technology Stack and Development Methodology

4.1 Development Methodology

The project follows the Agile software development methodology using iterative sprint-based implementation.

Each sprint includes:

- Sprint planning
- Task assignment
- Implementation
- Testing
- Sprint review
- Retrospective

This methodology allows continuous improvement and flexible adaptation to feedback.

4.2 Frontend Technologies

The frontend system is planned using:

- React
- Bootstrap
- Axios

React

React was selected because it enables modular and component-based UI development.

Bootstrap

Bootstrap simplifies responsive layout design and improves development speed.

Axios

Axios is used for communication between frontend and backend services.

4.3 Backend Technologies

The backend architecture is planned using:

- Node.js
- Express.js
- Sequelize ORM

Node.js and Express

Node.js and Express provide scalable REST API development.

Sequelize ORM

Sequelize simplifies database communication and model management.

4.4 Database

PostgreSQL is used as the primary relational database system.

The database stores:

- User profiles
- Study materials
- Forum posts
- Events
- Academic progress information

4.5 Authentication

JWT (JSON Web Token) authentication is used to:

- Secure login sessions
- Protect API routes
- Manage user access permissions

5. Stakeholder Analysis

Understanding stakeholders was important for identifying project expectations, benefits, and risks.

5.1 Student Body

Status

Positive

Expectations

- Fast access to study materials
- Better event visibility
- Easier collaboration with peers

Benefits

- Faster searching for materials
- Better student collaboration
- Stronger academic community

Possible Conflicts

- Spam and misinformation
- Privacy concerns

Team Response

- Content moderation
- Reporting systems
- Data protection measures

5.2 Student Organizations

Status

Positive

Expectations

- Easier promotion of activities
- Larger audience reach
- Better event engagement

Team Response

- Event posting tools
- Notification systems
- Event analytics

5.3 Professors

Status

Neutral

Expectations

- Productive student collaboration
- Protection of teaching materials
- Reliable content sharing

Possible Conflicts

- Sharing unauthorized materials
- Distribution of exam solutions

Team Response

- Content moderation tools
- Reporting copyrighted content
- Material removal requests

5.4 Tutors

Status

Negative

Concerns

- Reduced tutoring demand
- Sharing of paid materials

Team Response

- Promotion opportunities
- Proper content crediting
- Rules against copyrighted material sharing

5.5 Project Team

Status

Positive

Goals

- Successful project development
- Skill improvement
- Positive community impact

Challenges

- Limited resources
- Balancing stakeholder expectations
- Ongoing maintenance

6. Personas

To better understand target users, three personas were developed based on student needs, behaviors, and challenges.

6.1 Persona 1 – Ana Pećanac

Role

First-Year Student

Biography

Ana Pećanac is a 19-year-old first-year university student adjusting to independent academic life. She often feels overwhelmed by scattered study materials and uncertainty about exam preparation.

Main Goals

- Understand course material clearly
- Pass exams confidently
- Find reliable study resources
- Stay organized academically

Challenges

- Study materials are scattered across chats
- Difficulty identifying important topics
- Fear of asking questions publicly
- Difficulty tracking academic progress

Proposed Solutions

- Centralized repository of materials
- Progress tracking system
- Course-based discussion forum
- Verified and rated study resources

6.2 Persona 2 – Marko Jovanović

Role

Final-Year University Student

Biography

Marko Jovanović is a 23-year-old final-year student who mentors junior colleagues and actively participates in academic events.

Main Goals

- Organize study materials efficiently
- Help younger students
- Stay informed about events
- Prepare for professional career transition

Challenges

- Time wasted sharing materials repeatedly
- Repetitive mentoring questions
- Difficulty tracking relevant events

Proposed Solutions

- Centralized repository
- Forum with highlighted answers
- Event tracker and integrated calendar
- Deadline management system

6.3 Persona 3 – Sara Guberac

Role

Erasmus Student

Biography

Sara Guberac is a 21-year-old Erasmus student facing language barriers and difficulties integrating into a foreign academic environment.

Main Goals

- Access bilingual materials
- Connect with other students
- Stay informed about international events
- Manage deadlines effectively

Challenges

- Language barriers
- Limited peer connections
- Difficulty finding relevant events
- Managing academic obligations in a foreign system

Proposed Solutions

- Bilingual material uploads
- Erasmus-focused discussion categories
- Event filtering by language and type
- Progress tracking with multilingual support

7. Hypothesis Testing and Research

7.1 Initial Hypothesis

Students currently use multiple disconnected platforms for study materials, communication, and event tracking.

The hypothesis was:

Students want a centralized platform where they can access study materials, communicate with peers, track academic progress, and discover events.

7.2 Interview Process

Interviews were conducted with four students:

- Luka
- Sara
- Marko
- Nina

Participants came from different study programmes and academic years.

7.3 Main Findings

Study Material Problems

Students mainly relied on:

- Moodle
- WhatsApp groups
- Google Drive folders
- Discord servers

Most students complained that resources were disorganized.

Communication Needs

Students expressed interest in:

- Forums
- Peer support
- Quick question-and-answer systems

Academic Tracking Problems

Many students experienced confusion during exam periods and wanted:

- Better progress visibility
- Deadline tracking
- Centralized academic organization

Event Awareness

Students mainly discovered events through:

- Instagram
- Friends
- Posters
- Student organizations

They supported the idea of a centralized event tracker.

7.4 Research Conclusion

The research strongly supported the initial hypothesis.

The most requested features were:

- Study material sharing
- Discussion forums
- Academic progress tracking
- Event tracking

8. User Stories and Test Cases

User stories were developed to define system functionality from the perspective of real users.

8.1 Persona: Ana – First-Year Student

User Stories

- Access centralized course materials
- Track passed and pending exams
- Ask questions through forums

Example Test Cases

- Verify uploaded materials are displayed correctly
- Verify materials can be downloaded
- Verify forum questions can be posted
- Verify notifications are sent when questions receive answers

8.2 Persona: Marko – Senior Student

User Stories

- Upload study notes
- Highlight useful forum answers
- Track university events

Example Test Cases

- Verify multiple file formats are supported
- Verify contributors receive credit
- Verify events are listed chronologically
- Verify RSVP functionality works correctly

8.3 Persona: Sara – Erasmus Student

User Stories

- Access bilingual materials
- Follow classmates' activity
- Discover international events

Example Test Cases

- Verify materials can be filtered by language
- Verify English reminders are sent
- Verify users can follow and unfollow peers
- Verify events can be filtered by language

8.4 Importance of User Stories

User stories helped:

- Define functional requirements
- Understand user expectations
- Structure the testing process
- Prioritize platform features

9. Sprint 0 – Implementation Plan

Sprint 0 focused on preparing the development environment and organizing the project workflow.

9.1 Sprint 0 Goals

The team prepared:

- Technical architecture
- Development tools
- Product backlog
- Team responsibilities
- Initial workflow organization
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9.2 Development Tools

Version Control

- Git
- GitHub

Project Management

- Trello

Trello columns included:

- Backlog
- To Do
- In Progress
- Testing
- Done

Communication Tools

- Discord
- WhatsApp

Documentation Tools

- Google Docs
- Microsoft Office

9.3 Initial Product Backlog

Authentication System

- Registration
- Login
- JWT handling

Student Profile

- Faculty selection
- Programme selection
- Academic year setup

Study Materials Module

- Upload materials
- Browse materials
- Download materials

Progress Tracking Module

- Subject tracking
- Exam tracking

Discussion Forum

- Discussion posts
- Comment system

Event Tracker

- View events
- Submit events
- Approve events

Admin Dashboard

- User moderation
- Material moderation
- Event validation

9.4 Sprint Planning Overview

Phase 1

Core system implementation:

- Login system
- Profile management
- Materials module

Duration: 3 weeks

Phase 2

- Progress tracking
- Forum implementation

Duration: 3 weeks

Phase 3

- Event tracker
- Administrator dashboard

Duration: 2 weeks

10. System Architecture

The Student Hub Platform follows a three-layer architecture.

10.1 Presentation Layer

Implemented using React.

Responsibilities:

- User interface
- User interaction
- Displaying materials and events

10.2 Application Layer

Implemented using Node.js and Express.

Responsibilities:

- Business logic
- API communication
- Authentication
- Request handling

10.3 Data Layer

Implemented using PostgreSQL.

Responsibilities:

- Data storage
- User information management
- Forum data
- Event information
- Academic progress tracking

10.4 Benefits of Architecture

This architecture provides:

- Scalability
- Easier maintenance
- Better separation of concerns
- Improved development organization

11. User Testing Plan and Execution

Usability testing was conducted to evaluate how easily students could use the platform.

11.1 Testing Goals

The testing focused on:

- Registration and profile management
- Uploading and accessing materials
- Progress tracking
- Event navigation

11.2 Testing Method

Environment

- Laptop
- Chrome browser
- Stable internet connection

Testing Style

- Moderated usability testing
- Sessions lasting 25–30 minutes

Observed Metrics

- Time spent
- Errors
- User confusion
- Comments and suggestions

11.3 Test Participants

Participants:

- Marko
- Slobodan
- Djordje
- Jelena

Each participant represented a different type of student user.

11.4 Testing Tasks

Task 1 – Registration and Login

Users created accounts and configured profiles.

Task 2 – Uploading Materials

Users uploaded study files and assigned subjects.

Task 3 – Finding Materials

Users searched and downloaded materials.

Task 4 – Event Navigation

Users searched for events and viewed details.

Task 5 – Progress Tracking

Users updated subject completion status.

12. Testing Results and Analysis

12.1 Main Observations

Marko

- Fast and confident usage
- Slight confusion regarding allowed file formats

Slobodan

- Required guidance
- Navigation difficulties
- Confusion about required fields

Djordje

- Tested edge cases
- Attempted unauthorized profile editing
- Suggested easier correction flows

Jelena

- Requested better categorization
- Requested better event filtering
- Wanted clearer success confirmation messages

12.2 High Priority Issues

The following issues were identified as critical:

- Improve registration clarity
- Simplify navigation for beginners
- Add upload guidelines
- Add confirmation messages after actions

12.3 Medium Priority Issues

- Better event filtering
- Easier profile editing
- Simplified progress tracking interface

12.4 Suggested Improvements

- Add onboarding tutorials
- Add search suggestions
- Improve material categorization
- Add tooltips and guidance systems

12.5 Most Intuitive Feature

Finding and downloading materials was completed quickly by all users.

12.6 Least Intuitive Feature

Uploading materials caused the most confusion because users were unsure about:

- File requirements
- Navigation flow
- Allowed formats

12.7 Key Usability Insight

A major usability issue occurred when a tester entered incorrect profile information and expected to correct it without logging in.

The system correctly blocked unauthorized editing, but users expected:

- Clearer instructions
- Better error messages
- Easier correction workflows

13. Risks and Limitations

13.1 Technical Risks

Risks

- Deployment problems
- Server downtime
- Database connection issues

Mitigation

- CI/CD pipelines
- Backup hosting
- Regular testing

13.2 Resource Risks

Risks

- Limited time
- Small development team
- Tight deadlines

Mitigation

- Prioritizing MVP features
- Agile iteration planning
- Clear task distribution

13.3 Engagement Risks

Risks

- Low student participation
- Lack of user adoption

Mitigation

- Gamification features
- Faculty endorsement
- Better onboarding experience

13.4 Project Limitations

The project operates without a direct financial budget and relies on:

- Open-source technologies
- University hosting
- Student developer contributions

14. Recommendations and Future Improvements

Based on user testing and project analysis, several improvements are recommended for future development.

14.1 User Experience Improvements

- Add onboarding tutorials for first-time users
- Improve form validation and guidance
- Simplify navigation structure
- Improve upload workflow

14.2 Feature Improvements

- Advanced event filtering
- Better notification system
- Calendar integration
- Personalized recommendations

14.3 Accessibility Improvements

- Better multilingual support
- Improved accessibility for international students
- Responsive mobile optimization

14.4 Community Features

- Reputation systems
- Material ratings
- Mentor badges
- Gamification elements

14.5 Long-Term Development

Possible future extensions include:

- Mobile application development
- AI-powered study recommendations
- Integration with university systems
- Automated deadline reminders

15. Final Conclusion

The Student Hub Platform project successfully addressed the initial problem of fragmented academic resources and communication tools used by university students.

Through stakeholder analysis, hypothesis testing, personas, user stories, sprint planning, and usability testing, the team developed a strong conceptual foundation for a centralized student platform.

The research confirmed that students strongly value:

- Centralized study materials
- Easier peer communication
- Academic progress tracking
- Event organization systems

Usability testing demonstrated that the platform concept is intuitive for experienced users and highly valuable for student collaboration. However, the testing process also revealed several important areas for improvement, especially regarding onboarding, navigation clarity, and upload workflows.

Despite limitations related to time, manpower, and budget, the project achieved its primary goals and established a scalable foundation for future development.

The Student Hub Platform has strong potential to improve academic organization, student collaboration, and university engagement if fully implemented and expanded in the future.

Appendix

Main Planned Features

- Authentication System
- Student Profiles
- Study Material Repository
- Academic Progress Tracker
- Discussion Forums
- Event Management System
- Admin Dashboard

Core Technologies

- React
- Bootstrap
- Axios
- Node.js
- Express.js
- Sequelize ORM
- PostgreSQL
- JWT Authentication

Agile Workflow Summary

- Sprint Planning
- Iterative Development
- Continuous Testing
- Sprint Reviews
- Retrospectives